Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech
Code : 100606

Subject: Digital Signal Processing

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer the following (any seven question only):*** **[2 x 7 = 14]**

(a) Which of the following is a continuous-time signal?
- (i) Sinusoidal wave
- (ii) Impulse signal
- (iii) Unit step signal
- (iv) All of the above

(b) What does DTFT stand for?
- (i) Discrete-Time Fourier Transform
- (ii) Digital-Time Fourier Transform
- (iii) Dynamic-Time Fourier Transform
- (iv) Discrete Transfer Fourier Transform

(c) The process of converting continuous-time signals into discrete-time signals is called?
- (i) Quantization
- (ii) Sampling
- (iii) Aliasing
- (iv) Reconstruction

(d) What is the frequency domain representation of LTI systems called?
- (i) Fourier Series
- (ii) Fourier Transform
- (iii) Laplace Transform
- (iv) Both i and ii

(e) A system is causal if:
- (i) Output depends on future inputs
- (ii) Output depends only on present and past inputs
- (iii) Output is independent of input
- (iv) None of the above

(f) What is the result of convolving two impulse signals?
- (i) Impulse signal
- (ii) Zero signal
- (iii) Step signal
- (iv) Constant signal

(g) The sampling theorem states that to avoid aliasing, the sampling rate must be:
- (i) Less than the Nyquist rate
- (ii) Equal to the Nyquist rate
- (iii) Greater than or equal to twice the maximum signal frequency
- (iv) Greater than the Nyquist rate

(h) Which of the following is used to compute the convolution of two discrete signals?
- (i) Fourier Transform
- (ii) Laplace Transform
- (iii) Z-transform
- (iv) Convolution sum

(i) Which of the following represents the Nyquist rate?
- (i) Twice the highest frequency
- (ii) Half the highest frequency
- (iii) Equal to the highest frequency
- (iv) None of the above

(j) Which property of DFT leads to a reduction in computational complexity?
- (i) Symmetry
- (ii) Linearity
- (iii) Circular convolution
- (iv) Time-shifting

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Q.2 (a) Explain the advantages of DSP over analog signal processing. Provide practical examples to support your explanation. [7]

(b) Discuss the stability and causality conditions of discrete-time systems. Provide examples of systems that are stable and causal. [7]

Q.3 (a) Explain the basic elements of a DSP system with a block diagram. How does it function in practical applications? [7]

(b) Compare the correlation and convolution of discrete-time signals with examples. [7]

Q.4 (a) Differentiate between continuous-time and discrete-time signals. Use diagrams to illustrate their key characteristics. [7]

(b) Explain the properties of the Discrete-Time Fourier Transform (DTFT) for LTI system with applications. [7]

Q.5 (a) Derive the formula for the Inverse Discrete-Time Fourier Transform with suitable examples. [7]

(b) Explain the process of sampling continuous-time signals and the importance of the Nyquist rate. [7]

Q.6 (a) Discuss the relationship between the time-domain and frequency-domain representations of signals using the Fourier transform. [7]

(b) Explain the reconstruction of analog signals from their samples using an ideal low-pass filter. [7]

Q.7 (a) Explain the difference between FIR and IIR filters. What are the advantages and disadvantages of each? [7]

(b) Discuss the properties of the Discrete Fourier Transform (DFT). Explain its significance in digital signal processing. [7]

Q.8 (a) Derive the Fast Fourier Transform (FFT) algorithm and explain how it improves the computation of the DFT. [7]

(b) Determine the convolution for the two sequences $x(n) = \{3,2,1,2\}$, $(n) = \{1,2,1,2\}$ [7]

Q.9 (a) Illustrate how the properties of linearity and time-invariance simplify the analysis of discrete-time systems with examples. [7]

(b) Explain the concept of periodicity in continuous-time and discrete-time signals. How do their periodicity conditions differ? [7]

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